

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained module specifications, SW Unit test specifications, and test reports for assigned software modules, ensuring alignment with Software Architect guidelines and achieving **100%** compliance with project documentation standards.
- Authored production-ready source code in **Python** and **Java Spring Boot**, adhering strictly to coding guidelines and contributing to the development of new features for enterprise applications, resolving an average of **15** defects per release cycle.
- Integrated and tested software components, performing unit testing and peer reviews for code, design, and documentation, which reduced post-deployment critical issues by **20%**.
- Processed and pre-processed datasets from SMATA downloads by searching and identifying suitable traces, preparing data as instructed and contributing to the reduction of data preparation time by **30%** for machine learning model training.
- Executed training runs and experiments with various tuning parameters for neural networks, analyzing and summarizing results to inform model optimization strategies and contributing to a **15%** improvement in model accuracy.
- Managed artefacts within a **Configuration Management database**, committing code and documentation regularly to ensure version control integrity and support continuous integration pipelines, maintaining **99.9%** uptime for CI/CD processes.

SKILLS

Programming Languages – Python, C++, Java, R

Machine Learning & AI – Neural Rendering, Photorealism, Generative Models (NeRF, 3DGS), Model Training, Data Pre-processing, Computer Vision

Software Development – Module Specification, Unit Testing, Code Reviews, Clean Coding, Software Design Principles, REST APIs, Debugging

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

Tools & Technologies – SMATA, Git, Jenkins, Jira, Confluence, OpenGL ES, Vulkan

Data Analysis & Visualization – SQL, Power BI, DAX, Power Query, Tableau, pandas, NumPy

Domain Experience – Automotive Supplier Partner, Mobility Solutions, R&D;

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Neural Rendering Engine for Advanced 3D Visualisation (R&D; Project)

- Developed a novel **neural rendering** pipeline using **Python** and **PyTorch**, integrating techniques like NeRF to achieve photorealistic 3D scene reconstruction from sparse input data, reducing rendering time by **40%** compared to traditional methods.
- Engineered data pre-processing modules to prepare large-scale datasets for training generative models, implementing efficient data loading and augmentation strategies that improved training throughput by **25%**.
- Implemented GPU programming interfaces (simulated **OpenGL/Vulkan**) for accelerated inference, enabling real-time visualisation of complex 3D models and achieving frame rates suitable for interactive applications.

Automated Transaction Monitoring Dashboard (HCL Technologies)

- Designed and deployed a **payments operations dashboard** using **Java Spring Boot** and **PostgreSQL**, providing real-time monitoring of transaction flows and operational alerting for failed payments, improving incident detection time by **50%**.
- Automated data reconciliation workflows with **Python** scripts, cross-validating transaction records across multiple systems and reducing manual reconciliation efforts by **70%**.
- Implemented robust logging and monitoring with **Splunk** and **Dynatrace**, ensuring high availability and adherence to strict SLAs for financial transaction processing.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)